



DIGITAL DESIGN USING FPGA

-I2C “Slave” Project

-Omar Abdelmoneam Abdelkarim





Project Overview

I²C Master-Slave Communication System in Verilog HDL

This project presents the design and implementation of an Inter-Integrated Circuit (I²C) communication system using Verilog HDL. The primary objective is to develop a hardware-based I²C Master capable of communicating with external slave devices through the standard two-wire interface. I²C is a widely used synchronous serial protocol designed for short-distance communication between integrated circuits. It requires only two bidirectional lines: the Serial Clock Line (SCL), generated by the master, and the Serial Data Line (SDA), used for data transfer.

Current Implementation

- The project currently includes both the I²C Master and I²C Slave controllers, each implemented as a finite state machine (FSM). Together, they support the standard I²C communication sequence, including:
 - START and STOP condition detection/generation
 - 7-bit slave addressing with read/write control bit
 - ACK/NACK handling
 - Byte-level data transmission and reception
 - Slave memory read and write operations
- Supporting logic includes:
 - Counters and registers: Bit counters track transferred bits, while address and data registers store the received address and transmitted/received data.
 - Tri-state drivers: The SDA line is controlled using tri-state logic to implement the open-drain behavior required by the I²C protocol.
- Slave memory: The slave contains an internal memory used to store received data and provide data during read operations.

Design Approach

- Modular structure: The Master, Slave, FSMs, counters, memory, and SDA/SCL control logic are implemented as separate functional blocks for easier design and debugging.
- Simulation and verification: Individual modules have been tested using dedicated Verilog testbenches to verify correct protocol operation.
- Scalability: The modular architecture allows straightforward integration of the Master and Slave into a complete I²C communication system.

Future Work

- Complete the I²C Slave testbench.
- Develop the top-level module to integrate the I²C Master and Slave.
- Verify complete Master-Slave communication through simulation and FPGA implementation



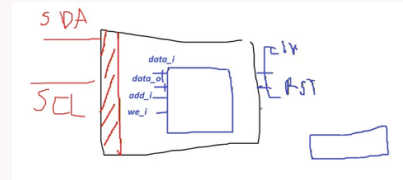
Resources

- <https://www.youtube.com/watch?v=S61SF7YHTdA>
- <https://www.analog.com/en/resources/technical-articles/i2c-primer-what-is-i2c-part-1.html>
- **NTI slides provided by the instructor**

Code Summary

The I²C Slave was implemented by dividing the design into several functional blocks, each responsible for handling a specific part of the communication with the I²C master.

Ports



- **State Machine (FSM)**

The controller is built around a finite state machine (FSM) that manages the slave operation. The FSM transitions through the IDLE, ADDRESS, ACK_ADDR, WRITE_DATA, ACK_WRITE, READ_DATA, and ACK_READ states to receive the address, perform read or write operations, and complete the transaction.

- **Start and Stop Detection**

The slave continuously monitors the SDA line to detect the start and stop conditions. These signals determine when a transaction begins and ends, allowing the FSM to enter or exit the communication process.

- **Address Recognition**

The received address byte is stored and compared with the programmed slave address. If the addresses match, the slave generates an ACK and proceeds with either a read or write operation according to the read/write bit.

- **Data Handling**

During write operations, incoming data is received serially and stored in the slave memory. During read operations, data is fetched from the memory and transmitted serially to the master.

- **Bit Counter and Memory**

A 4-bit counter tracks the number of bits transferred during each byte. The slave also includes an internal 256-byte memory used to store received data and provide data during read operations.

- **SDA Control**

The SDA line is controlled using tri-state logic to implement the open-drain behavior of the I²C bus. The slave drives SDA only when transmitting data or acknowledgment bits and releases the line during data reception.

No Errors in Lint or Vivado

Lint no errors and schematic

The screenshot shows the Vivado IDE with the I2C_Slave.v Verilog file open. The code defines an I2C slave module with inputs for clock, reset, write enable, slave address, read data, and SCL. It includes registers for slave memory, SDA, SCL, and address matching. The schematic view on the right shows the hardware implementation of the module.

Lint Checks

Severity	Status	Check	Alias	Message	Module	Category	State	Owner	STARC Reference
Warning	Open	async_reset_active_high		Asynchronous reset is active high. Reset rst, Module I2C...	I2C_Slave	Clock	open	unassign...	2.3.6.2
Warning	Open	inout_port_exists		Inout port exists in the design. Signal SDA, Module I2C...	I2C_Slave	Connectivity	open	unassign...	
Warning	Open	inout_port_exists		Inout port exists in the design. Signal SCL, Module I2C...	I2C_Slave	Connectivity	open	unassign...	
Warning	Open	always_signal_assign_large		Always block has more signal assignments than the spe...	I2C_Slave	Rtl Design Style	open	unassign...	2.6.1.3
Warning	Open	comment_not_in_english		Comments include non-English characters. File C/User...	none	Nomenclature ...	open	unassign...	3.5.6.4, 3.5.6.6
Warning	Open	comment_not_in_english		Comments include non-English characters. File C/User...	none	Nomenclature ...	open	unassign...	3.5.6.4, 3.5.6.6
Warning	Open	comment_not_in_english		Comments include non-English characters. File C/User...	none	Nomenclature ...	open	unassign...	3.5.6.4, 3.5.6.6
Warning	Open	multi_ports_in_single_line		Multiple ports are declared in one line. Module I2C_Sla...	I2C_Slave	Rtl Design Style	open	unassign...	3.5.6.3

Vivado no errors and schematic

The screenshot shows the Vivado Messages window with the 'Messages' tab selected. It displays a list of messages categorized by Vivado Commands, Elaborated Design, Synthesis, Synthesized Design, Implementation, and Implemented Design. The 'Implemented Design' category is highlighted, showing 2 warnings and 5 infos.

ELABORATED DESIGN - xc7w4100f1927-2 (active)

The Schematic view shows the hardware implementation of the I2C_Slave module, with a green background and various logic components.